



## Regular Article

## Artificial intelligence, governance, and economic growth: Toward a new theory of digital developmentalism

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## ABSTRACT

This paper develops a new theoretical framework titled Digital Developmentalism, which explains how governments in the age of artificial intelligence (AI) and data-driven technologies are reshaping the mechanisms of economic growth and structural transformation. Integrating insights from developmental state theory, institutional economics, and general-purpose technology (GPT) theory, the study conceptualizes AI not merely as a technological innovation but as a transformative governance capability. The framework positions the state as a computational orchestrator, leveraging algorithmic coordination, data infrastructures, and public-private digital ecosystems to guide innovation and policy learning. Four key mechanisms underpin digital developmentalism: (1) algorithmic coordination, (2) data as developmental infrastructure, (3) AI-enhanced policy feedback, and (4) public-private digital synergy. Together, these mechanisms explain how AI governance capacity enables governments to manage complexity, enhance productivity, and foster inclusive development. The study also explores ethical and political challenges such as algorithmic opacity, surveillance, and inequality, emphasizing the need for transparent, accountable, and human-centered AI governance. By uniting governance theory and GPT theory into a single conceptual model, this paper contributes to the emerging discourse on how digital technologies redefine state capacity and the future of development. It concludes by proposing pathways for empirical testing and comparative research on AI-driven governance and economic performance across different national contexts.

## 1. Introduction

The rapid diffusion of artificial intelligence (AI) across governments, markets, and societies is fundamentally reshaping the relationship between governance and economic growth. As a general-purpose technology, AI permeates multiple sectors, from manufacturing and finance to health and education, while simultaneously transforming how governments design, implement, and evaluate policy (Agrawal et al., 2024; Brynjolfsson & McAfee, 2014). Governments now use AI to enhance administrative efficiency, forecast economic trends, optimize resource allocation, and support innovation-driven growth strategies. These shifts mark a new phase in the political economy of development, where the integration of AI and data-driven decision systems creates unprecedented capacities for coordination and transformation.

Yet, despite the ubiquity of AI in policy discourse, the theoretical understanding of how AI-enabled governance contributes to long-run economic development remains underdeveloped. Traditional frameworks of economic growth, from neoclassical to institutional and developmental state theories, have yet to capture the distinctive

mechanisms by which AI reshapes state capacity, industrial upgrading, and innovation ecosystems (Alshahrani et al., 2024; Yerlikaya & Erzurumlu, 2021). In this paper, digital developmentalism is defined as a governance paradigm in which the state strategically leverages artificial intelligence, data, and digital platforms to coordinate innovation, guide structural transformation, and promote long-term economic growth.

The fundamental problem lies in the widening gap between technological change and developmental theory. While the 20th century developmental state model emphasized bureaucratic autonomy, industrial policy, and strategic coordination between government and business (Evans, 1995; Johnson, 1982), the 21st century digital state operates through algorithmic coordination, real-time data analytics, and adaptive policy systems. This transition suggests that the mechanisms through which governments stimulate growth have evolved from hierarchical planning to networked, data-driven steering. However, much of the existing research on digital governance focuses on administrative efficiency and service delivery (Hossin et al., 2023), rather than on the broader macroeconomic transformation that AI may induce. As such, the

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question remains: how does AI governance translate into developmental outcomes, and what theoretical framework can capture this new configuration of state–market interaction? Against this backdrop, the primary objective of this study is to develop a new theoretical framework, digital developmentalism, that explains how artificial intelligence-enabled governance reshapes state capacity and economic growth in the digital era. Specifically, the paper asks how AI-driven governance mechanisms transform the traditional relationship between the state, innovation, and structural economic development.

This study argues that the current era calls for a reconceptualization of state-led development, a shift from classical developmentalism to what may be termed digital developmentalism. Digital developmentalism refers to a governance paradigm in which the state leverages AI, big data, and digital infrastructures to steer innovation, enhance productivity, and coordinate economic transformation. In this framework, AI is not merely a technological input but a governance instrument that augments state capacity and institutional learning. By embedding predictive analytics and machine learning into public management systems, governments acquire new forms of intelligence and agility that enable continuous policy calibration (OECD, 2023; Zhou et al., 2024). This adaptive capability differentiates digital developmentalism from earlier forms of state-led modernization, where policy interventions were largely reactive and information-poor. The emerging AI-driven state possesses the ability to learn from data, anticipate shocks, and allocate resources in real time, attributes that redefine the logic of developmental governance. The theory developed in this paper advances four interrelated mechanisms, algorithmic coordination, data as developmental infrastructure, AI-enhanced policy feedback, and public–private digital synergy, that together explain how AI-enabled governance reshapes state capacity and economic growth.

The theoretical and empirical landscape highlights the need for a new synthesis. Classical developmental theories explain growth through industrial policy, human capital, and export competitiveness (Amsden, 1989; Wade, 2018). Institutional economics emphasizes the role of property rights, rule of law, and state capacity (Acemoglu & Robinson, 2013). Yet neither tradition accounts for algorithmic governance capabilities, the state's ability to process data, model complexity, and predict economic behavior. In parallel, contemporary digital governance research provides valuable insights into AI adoption in public administration but remains disconnected from macroeconomic theory (Van Noordt & Misuraca, 2022). By integrating these strands, the concept of digital developmentalism aims to explain how the fusion of AI technologies and governance capacities shapes economic growth trajectories in the digital age.

The argument advanced in this study proceeds from the assumption that the state remains a central actor in development but that its capabilities are being reconstituted by digital technologies. AI enhances the informational and analytical dimensions of state capacity, enabling governments to act as algorithmic coordinators of complex socio-economic systems (Mergel et al., 2024). For example, AI-driven platforms now facilitate targeted industrial policies, evidence-based fiscal planning, and real-time monitoring of supply chains, functions that were previously constrained by informational bottlenecks. This transformation implies that the effectiveness of government in promoting economic growth increasingly depends on its AI governance capacity: the institutional ability to harness data, algorithms, and digital ecosystems for developmental purposes (World Bank, 2024). The digital developmental state thus emerges as an adaptive, learning-oriented entity that governs not only through regulation but also through computation and prediction.

This study develops a theoretical framework that conceptualizes the relationship between AI, governance, and economic growth through the lens of digital developmentalism. The objective is not to empirically measure these relationships but to articulate a conceptual model that captures the causal and institutional mechanisms linking AI-driven governance capabilities to developmental outcomes. The framework

identifies key components, algorithmic coordination, data infrastructure, policy feedback loops, and public–private digital ecosystems, as the building blocks of this new developmental logic. By doing so, it seeks to bridge the gap between governance theory, innovation economics, and digital transformation studies, offering an integrative approach to understanding economic modernization in the AI era.

This study pursues two analytically distinct aims. Its primary aim is descriptive and explanatory: to develop a theoretical framework that explains how AI-enabled governance mechanisms reshape state capacity and economic growth. Normative considerations, such as ethical governance, inclusiveness, and democratic accountability, are treated as implications that arise from the analytical framework rather than as prescriptive assumptions embedded within it. By maintaining this distinction, the paper seeks to clarify what digital developmentalism explains and how its insights may later inform normative and policy debates. Methodologically, this study adopts a theory-building conceptual synthesis approach. Rather than conducting a systematic or meta-analytic literature review, the paper integrates and synthesizes key theoretical insights from developmental state theory, institutional economics, general-purpose technology theory, and digital governance research. This integrative approach enables the identification of common mechanisms, conceptual gaps, and complementary logics across literatures, which are then recombined into a coherent framework explaining how AI-enabled governance reshapes state capacity and economic growth.

## 2. Literature review: from classical developmentalism to the digital state

This section develops the theoretical foundations linking artificial intelligence, governance capacity, and economic growth, drawing on institutional and evolutionary perspectives.

### 2.1. Classical developmental state paradigm

The developmental state theory emerged as one of the most influential explanations for the economic transformation of East Asia during the late twentieth century. Originating from comparative studies of Japan, South Korea, and Taiwan, it provided a framework for understanding how proactive and capable governments can guide industrialization and technological upgrading (Amsden, 1989; Evans, 1995; Johnson, 1982). In these contexts, governments did not merely correct market failures; rather, they actively shaped markets through industrial policy, targeted subsidies, export promotion, and long-term planning. Bureaucratic autonomy, coupled with meritocratic recruitment and technocratic expertise, allowed states to implement complex economic strategies insulated from short-term political pressures. This combination of autonomy and embeddedness, what Evans (1995) termed “embedded autonomy”, enabled effective coordination between the state and private sector actors, balancing state direction with business cooperation.

The developmental state's success lay in its institutional design. Ministries such as Japan's ministry of international trade and industry or Korea's economic planning board acted as “pilot agencies,” orchestrating national development priorities and managing industrial upgrading (Kim et al., 2022). Strategic collaboration between the state and large conglomerates allowed for capital accumulation, technological transfer, and productivity growth. The outcomes were dramatic: rapid industrialization, export-led growth, and structural transformation from agrarian to industrial economies (Wade, 2018). However, this model was rooted in an industrial-era paradigm, where the key drivers of growth were physical capital, manufacturing capacity, and labor productivity. Its institutional logic relied on hierarchical coordination, stable policy horizons, and relatively slow technological diffusion, conditions that differ markedly from the fluid, data-intensive environment of the digital age. As contemporary economies become

more knowledge- and platform-based, the developmental state model requires theoretical renewal to account for new forms of digital coordination and AI-driven governance (Song et al., 2025).

## 2.2. Institutional and governance approaches to growth

Following the developmental state tradition, institutional economics expanded the analysis of growth by emphasizing the rules, norms, and governance structures that underpin economic performance. North (1990) defined institutions as “the rules of the game” that structure human interaction, shaping incentives for innovation, investment, and exchange. Later, Acemoglu and Robinson (2012) differentiated between inclusive institutions, those that foster participation and innovation, and extractive ones that concentrate power and stifle productivity. From this perspective, economic growth results not only from capital accumulation but from institutional arrangements that enable efficient coordination, protect property rights, and encourage entrepreneurship.

Parallel to institutional economics, governance research emphasized the quality and effectiveness of public administration as a determinant of development outcomes. Kaufmann et al. (2011) operationalized governance through indicators such as government effectiveness, rule of law, regulatory quality, and control of corruption, demonstrating strong correlations with GDP per capita. These frameworks positioned governance capacity as a key enabler of sustained economic development. However, emerging critiques suggest that traditional measures of governance fail to capture the new capabilities required in a digital economy (Medaglia et al., 2024; OECD, 2023). Institutional quality is no longer only about bureaucratic efficiency or legal stability; it increasingly depends on a government's capacity to manage and deploy data, algorithms, and digital infrastructures. Governance today involves not only human decision-making but also computational intelligence, automated systems, and cross-sectoral digital networks (Janssen et al., 2022). Consequently, institutional economics must evolve to include digital capacity, the ability of institutions to harness AI and data-driven tools for coordination, foresight, and adaptation.

## 2.3. Rise of digital governance and smart states

Over the past two decades, the evolution of governance has mirrored broader technological transformations, from the early phase of e-government, focused on service delivery, to digital government, emphasizing integration and citizen engagement, and now to AI-driven governance, which utilizes predictive analytics and automation to support decision-making (Milakovich, 2021). This progression marks a qualitative shift in how the state operates: from process efficiency to algorithmic intelligence. The contemporary “smart state” is characterized by the use of data platforms, machine learning, and algorithmic regulation to enhance transparency, forecast risks, and optimize policy interventions (de Jong et al., 2025). Governments increasingly act as data platforms, coordinating across ministries and sectors through interoperable systems that enable real-time policy monitoring and adaptive management.

Illustrative cases highlight how AI governance is redefining state capabilities. Estonia's X-Road platform provides a backbone for data exchange across public and private institutions, supporting predictive policy applications and digital trust. Singapore's Smart Nation initiative integrates AI, sensors, and analytics to optimize transport, urban planning, and healthcare systems (UNDP, 2024). South Korea's National AI Strategy 2030 aims to embed AI across industrial and administrative domains to accelerate productivity and public value creation (Park & Chang, 2022). These initiatives reveal how digital infrastructures function as new forms of developmental capital, enhancing the informational capacity of the state and enabling adaptive, evidence-based policy-making. AI thus transforms governance from a static structure into a dynamic ecosystem that continuously learns, predicts, and coordinates.

## 2.4. Gaps in the literature

Despite growing recognition of AI's role in government, the theoretical integration of governance, technology, and development remains incomplete. The digital governance literature primarily emphasizes service efficiency, transparency, and citizen engagement (Hossin et al., 2023; Milakovich, 2021), while macroeconomic research continues to rely on industrial-era frameworks that overlook data and algorithmic infrastructures as productive assets. Similarly, developmental state theory, though historically successful in explaining industrial transformation, remains tied to analog models of planning, bureaucracy, and industrial policy (Sonn & Choi, 2022). What is missing is a comprehensive theoretical framework that captures how AI-enabled governance capacities influence innovation, productivity, and structural change at the systemic level.

This gap underscores the need for a new synthesis between governance theory, innovation economics, and digital technology. The notion of digital developmentalism provides such an integrative approach by linking AI governance capacity to economic performance. It argues that data, algorithms, and digital ecosystems now constitute critical factors of production, comparable in importance to capital and labor in previous eras. The state's developmental role thus extends beyond regulation or industrial policy to include the orchestration of digital infrastructures and algorithmic coordination mechanisms. Addressing this theoretical vacuum is essential for understanding how governments can harness AI not only to improve efficiency but also to drive sustained, inclusive, and innovation-led growth.

## 3. Theoretical background: linking AI, governance, and growth

### 3.1. Institutional and evolutionary perspectives

The institutional and evolutionary traditions in economic theory provide critical foundations for understanding the dynamic relationship between governance, technology, and growth. North's (1990) seminal work defined institutions as “the rules of the game”, both formal and informal constraints that shape human interaction and economic performance. Institutions, in this sense, determine the incentives for innovation, investment, and productive efficiency by structuring the way economic actors coordinate and compete. Stable and predictable institutional environments lower transaction costs, foster trust, and facilitate the diffusion of technologies, thereby enhancing long-term productivity growth. North's insight remains pivotal: technological progress does not occur in a vacuum but is embedded in institutional arrangements that either enable or inhibit adaptation. However, as the pace of technological change accelerates in the digital era, institutions must evolve from being static frameworks to adaptive systems capable of learning and transformation.

Evolutionary economists extended this view by emphasizing that innovation and growth are path-dependent, cumulative, and non-equilibrium processes (Nelson & Winter, 1985). Economic development unfolds as a series of technological paradigms shaped by routines, learning-by-doing, and feedback between firms, institutions, and policy environments. Unlike neoclassical models that assume equilibrium and perfect information, evolutionary approaches treat economic systems as constantly adapting to new knowledge and uncertainty. In this framework, governments play a critical role in nurturing innovation ecosystems, supporting experimentation, and creating enabling conditions for technological diffusion. The key insight for AI-era development is that governance itself must evolve as part of this adaptive process. Just as firms innovate to remain competitive, states must innovate institutionally to remain effective in coordinating complex, fast-changing socio-economic systems (Birkstedt et al., 2023).

Digital technologies, particularly AI, introduce new evolutionary pressures that challenge the traditional modes of governance and policymaking. Institutions are now required to process vast amounts of real-

time data, respond to non-linear feedback, and regulate technologies whose social impacts are uncertain and emergent. This environment demands adaptive governance, institutions that can learn, recalibrate, and evolve based on continuous information flows (Schiff, 2023). In contrast to the bureaucratic rigidity of the classical developmental state, the digital state operates as a dynamic learning system. The evolutionary logic thus extends beyond firms to governments, suggesting that governance capabilities themselves are subject to evolutionary selection: those governments that effectively integrate AI and digital tools into policymaking are better positioned to sustain innovation-led growth and resilience. This synthesis of institutional and evolutionary perspectives provides a foundation for theorizing how AI transforms the mechanisms of state capacity and economic coordination.

### 3.2. AI as a general-purpose technology

AI represents a paradigmatic case of a general-purpose technology, a class of innovations that transform economies through pervasive, complementary, and long-term productivity effects (Bresnahan & Trajtenberg, 1995). Like steam power, electricity, and information technology, AI exhibits three defining characteristics: wide applicability across sectors, potential for continuous improvement, and capacity to spawn complementary innovations (Brynjolfsson & McAfee, 2014). However, AI is distinct from prior GPTs in one crucial respect: while earlier technologies automated physical tasks, AI automates cognitive and coordination functions. It enhances the capacity to perceive, predict, and decide, core elements of governance and economic organization. Consequently, AI's general-purpose nature extends beyond firms and markets to encompass the public sector, where decision-making, regulation, and policy design are increasingly shaped by algorithmic processes (Uuk et al., 2025).

From an economic standpoint, AI's productivity effects are mediated through its role as an information coordination technology. By transforming how information is collected, analyzed, and acted upon, AI changes the structure of decision-making and governance. It enables governments to perform three key functions more effectively: real-time monitoring, predictive regulation, and dynamic resource allocation. Through real-time data analytics, governments can detect economic anomalies, forecast unemployment, or anticipate infrastructure bottlenecks before they occur (World Bank, 2024). Predictive regulation allows policymakers to model risk scenarios, optimize compliance mechanisms, and identify systemic vulnerabilities (Alshahrani et al., 2024). Finally, AI-driven resource allocation, through automated budget modeling or smart procurement, permits the fine-tuning of fiscal policy and investment priorities in response to changing conditions. These capabilities fundamentally alter the state's coordination role, positioning governments as algorithmic coordinators rather than hierarchical planners.

This transformation carries deep theoretical implications. If the industrial-era developmental state was characterized by bureaucratic intelligence and industrial coordination, the AI-era state is characterized by computational intelligence and algorithmic coordination. Governments now manage complexity not through bureaucratic expansion but through data architectures and machine learning systems that scale decision-making capacity. In this sense, AI functions as an extension of governance cognition, an augmentation of institutional intelligence (Zhou et al., 2024). Moreover, AI's diffusion creates new complementarities between public and private sectors, as innovation increasingly occurs within digital ecosystems co-governed by data standards, platform infrastructures, and algorithmic governance frameworks. Thus, AI as a GPT redefines the boundaries of the developmental state by embedding governance within technological networks that continuously produce, process, and apply knowledge for economic transformation.

### 3.3. Governance capability and economic outcomes

The effectiveness of government in fostering economic growth depends on what may be termed governance capability, the ability of state institutions to steer economic activity through coherent policies, strategic coordination, and adaptive regulation. Traditionally, governance capability has been analyzed in terms of bureaucratic quality, policy coherence, and administrative efficiency (Kaufmann et al., 2011). In the digital era, however, such definitions are incomplete. Economic competitiveness increasingly hinges on the state's ability to manage information flows, deploy data infrastructures, and integrate AI systems into decision-making processes. This shift calls for a new construct, AI governance capability (AIGC).

AIGC can be defined as the institutional capacity to harness data-driven technologies, particularly AI, for policy formulation, implementation, and evaluation (Valle-Cruz et al., 2020). It encompasses three interrelated dimensions. First, data infrastructure: the foundational layer of interoperable databases, digital identity systems, and connectivity that enables information exchange and analysis. Second, algorithmic decision systems: the tools and platforms that apply machine learning to governance tasks such as predictive analytics, policy simulation, and automated regulation. Third, institutional learning: the adaptive processes through which governments interpret algorithmic outputs, refine policies, and integrate feedback into long-term planning (de Jong et al., 2025; OECD, 2023). Together, these dimensions transform governance from a rule-based to a learning-based enterprise.

The theoretical hypothesis proposed here is that AI Governance Capability enhances the effectiveness of state-led growth strategies by improving the state's informational precision, responsiveness, and coordination efficiency. Governments endowed with high AIGC can better identify emerging opportunities, mitigate systemic risks, and facilitate innovation diffusion across sectors. This, in turn, strengthens the link between governance quality and economic performance, a relationship well established in traditional development theory but now mediated through digital mechanisms (Rodrik & Stiglitz, 2025). AIGC thus represents an evolution of state capacity, where policy effectiveness is determined not only by institutional integrity but also by computational intelligence and data-driven agility.

In the broader developmental context, AIGC redefines the nature of economic policymaking. It shifts the focus from industrial targets and sectoral interventions to the creation of digital infrastructures and algorithmic ecosystems that underpin innovation and productivity. For developing economies, this offers a pathway to leapfrog traditional stages of industrialization by investing in AI capabilities that enhance coordination and competitiveness (UNDP, 2024). Yet, it also introduces new governance challenges, including ethical oversight, data privacy, and algorithmic bias, which must be managed to ensure that digital transformation remains inclusive and equitable (Cowls et al., 2023). Thus, the capacity to govern AI effectively becomes not only an economic asset but a determinant of developmental legitimacy.

The core concepts advanced in this study were developed through a process of conceptual extension and theoretical synthesis rather than inductive empirical generalization. The construct of AI Governance Capability was derived by extending established notions of state capacity and governance effectiveness to account for the role of data infrastructures, algorithmic decision systems, and institutional learning emphasized in contemporary digital governance and AI policy literatures. Similarly, digital developmentalism was formulated as a conceptual extension of classical developmental state theory, adapted to reflect the informational, computational, and coordination dynamics introduced by artificial intelligence. Rather than introducing entirely novel constructs, the study recombines and refines existing theoretical elements to explain governance transformation under conditions of digital and AI-driven change.

### 3.3.1. Methodological note: framework-building procedure

Framework development followed a structured conceptual synthesis process. First, relevant theoretical literatures were identified based on their analytical relevance to state capacity, governance mechanisms, and economic growth under conditions of technological change. Second, key concepts and mechanisms were extracted from each literature stream, with particular attention to coordination, information processing, institutional learning, and state–market interaction. Third, overlapping and complementary mechanisms were systematically compared across literatures to identify convergent themes and conceptual gaps. Finally, these mechanisms were refined and recombined into an integrated framework, digital developmentalism, ensuring internal coherence, theoretical consistency, and clear analytical boundaries. This iterative comparison and refinement process guided the selection of the four core mechanisms and their positioning within the conceptual model.

In the absence of statistical or empirical methods, the analytical strategy of this study relies on deductive theoretical reasoning and comparative conceptual synthesis. Propositions regarding AI governance and economic growth are generated by systematically comparing established theoretical mechanisms across developmental state theory, institutional economics, and digital governance scholarship. Through this deductive process, the study identifies how AI-enabled coordination, data infrastructure, and policy learning logically extend existing explanations of state-led development under conditions of technological change. This form of analytical reasoning enables the derivation of theoretically grounded propositions without relying on empirical estimation.

## 4. Conceptualizing digital developmentalism

This section introduces digital developmentalism as a theoretical framework and specifies its core mechanisms, propositions, and conceptual architecture.

### 4.1. Definition and core idea

Digital developmentalism represents a new governance paradigm in which the state strategically employs AI, data infrastructures, and digital platforms to steer innovation, enhance productivity, and drive structural economic transformation. It extends the foundational logic of the classical developmental state, mission-oriented governance, long-term planning, and state–market coordination, into the digital era, where coordination occurs not through bureaucratic hierarchies but through algorithmic systems and data-driven decision architectures (Mazzucato, 2021). The concept rests on the recognition that digital technologies are not merely tools of administration but fundamental determinants of developmental capacity. In this framework, the state becomes both a regulator and an orchestrator of digital ecosystems, guiding the flow of data, capital, and innovation toward national development objectives.

Whereas traditional developmentalism relied on ministries, planning commissions, and industrial policy agencies as instruments of coordination, digital developmentalism operates through digital governance mechanisms, interconnected databases, AI models, and platform infrastructures that enable real-time monitoring, policy experimentation, and adaptive learning (Bresser-Pereira, 2023). The essence of this paradigm is mission-oriented digital coordination: the state sets strategic goals, such as green transition or AI-driven competitiveness, and deploys algorithmic tools to mobilize stakeholders, allocate resources, and measure progress dynamically. This approach aligns with the shift in global development thinking from static industrial upgrading to dynamic innovation ecosystems, where digital capabilities are integral to competitiveness and resilience (Khan & Christiansen, 2011).

Digital developmentalism thus redefines the state's developmental role along three dimensions. First, temporal, by transforming governance from reactive to predictive; second, spatial, by decentralizing

coordination through interconnected digital systems; and third, informational, by turning data into a core developmental asset. It proposes that economic transformation in the twenty-first century depends on how effectively governments embed AI into the logic of policymaking, regulation, and collaboration. The result is a state that is both intelligent, in its capacity to analyze and anticipate, and adaptive, in its ability to learn and evolve through feedback.

### 4.2. Four mechanisms of digital developmentalism

Digital developmentalism operates through four interrelated mechanisms that together define the architecture of AI-enabled state capacity: algorithmic coordination, data as developmental infrastructure, AI-enhanced policy feedback, and public–private digital synergy. Each mechanism represents a distinct function through which AI reshapes the developmental process.

#### a) Algorithmic Coordination

Algorithmic coordination refers to the use of AI systems to manage economic processes through predictive analytics, automated regulation, and data-driven planning. Unlike bureaucratic coordination, which depends on hierarchical command structures and ex post information, algorithmic coordination enables real-time responsiveness and anticipatory governance (Van Noordt & Misuraca, 2022). Governments increasingly deploy AI for tasks such as forecasting tax revenues, detecting fraud in public procurement, optimizing energy grids, or modeling labor-market dynamics. For instance, Singapore's Smart Nation initiative employs machine learning models to monitor transportation efficiency and urban resource allocation, exemplifying the state's role as a computational coordinator of complex systems (Yeo, 2023).

Theoretically, algorithmic coordination transforms the state's informational function. It reduces asymmetries between public and private actors by integrating multi-source data, thereby enhancing collective intelligence. This mechanism aligns with the dynamic capabilities perspective in organizational theory: governments with stronger algorithmic coordination can sense opportunities, seize them through timely intervention, and reconfigure resources as conditions change. In doing so, they replicate the developmental state's strategic guidance function, but with exponentially greater speed and granularity.

#### b) Data as Developmental Infrastructure

In digital developmentalism, data is the new developmental infrastructure, a resource that underpins innovation, policy learning, and economic competitiveness. Just as physical infrastructure, such as roads, energy, and telecommunications, enabled industrialization, digital infrastructure, data networks, cloud platforms, and interoperable systems, enables economic modernization in the AI age (World Bank, 2024). Data serves dual roles: as an input for innovation (fueling AI models and analytics) and as an output of governance (producing continuous feedback on socio-economic conditions). The accumulation and structuring of public data thus become acts of economic policy, determining the quality of insights available for both public and private innovation.

Governments increasingly treat data as a public good that can catalyze ecosystem-wide productivity. For example, Estonia's X-Road platform facilitates secure data sharing among institutions, while Korea's open-data policy integrates information from over 900 public agencies to stimulate AI entrepreneurship (OECD, 2023). However, managing data as a developmental asset also entails building trust through ethical governance, privacy protection, and transparency (Wirtz et al., 2019). In this sense, digital developmentalism requires institutional frameworks that ensure data sovereignty and responsible innovation, balancing economic utility with citizen rights. By institutionalizing data as public

capital, governments create the informational foundation necessary for sustained innovation-led growth.

c) AI-Enhanced Policy Feedback

A defining feature of digital developmentalism is the transformation of policy feedback loops through AI. Traditional policy cycle, agenda setting, formulation, implementation, evaluation, often suffered from lagged information and limited learning capacity. AI changes this dynamic by enabling continuous feedback, where policies are monitored, evaluated, and recalibrated in real time (Alshahrani et al., 2024). Machine learning models can detect unintended consequences, simulate alternative scenarios, and optimize policy interventions iteratively. For instance, predictive analytics in social welfare systems can identify emerging poverty clusters and adjust benefits distribution dynamically (UNDP, 2024).

This AI-enhanced feedback creates a form of reflexive governance, where decision-making is constantly informed by updated evidence. The state thus evolves into a learning system, capable of policy adaptation through algorithmic experimentation. The implications for developmental theory are profound: while the classical developmental state was guided by ex-ante industrial plans, the digital developmental state operates through adaptive planning, where AI continuously refines developmental pathways. This mechanism reinforces governance resilience, allowing the state to respond not only to market volatility but also to technological disruption, pandemics, and environmental change. In effect, AI transforms governance from an administrative apparatus into an intelligent organism, capable of sensing, analyzing, and evolving alongside the economy it governs.

To integrate these traditions into a unified framework, Table 1 summarizes the core theoretical anchors that inform the concept of digital developmentalism, indicating how each lens contributes to understanding the nexus between AI, governance, and economic growth.

d) Public-Private Digital Synergy

The fourth mechanism, public-private digital synergy, concerns the collaborative interface between state and market actors within digital ecosystems. In digital developmentalism, the state is neither a passive regulator nor a command planner; it is a platform orchestrator that fosters co-creation among private firms, startups, academia, and civil society (Mazzucato, 2021). Through shared digital infrastructures, such as AI sandboxes, open APIs, and innovation hubs, the government creates enabling environments where public missions and private capabilities align. This synergy accelerates innovation diffusion, enhances digital entrepreneurship, and scales technological solutions across industries.

Examples of this mechanism are evident in Finland's AI Business program, Singapore's GovTech partnerships, and China's state-enterprise collaborations in AI governance. Each illustrates how digital ecosystems operate as developmental coalitions, where public goals,

such as sustainability, inclusion, and resilience, are advanced through collaborative innovation. This mechanism also highlights the networked nature of the digital developmental state: economic growth emerges not from centralized control but from the coordination of distributed intelligence. Effective public-private digital synergy requires institutional trust, regulatory agility, and interoperability standards, conditions that determine whether collaboration yields equitable developmental outcomes or reinforces monopolistic power (Balarabe, 2025).

4.3. Theoretical propositions

The propositions are analytically derived and theory-driven; they articulate hypothesized mechanisms linking AI-enabled governance and economic outcomes rather than empirically validated causal relationships. These propositions do not assume that AI adoption automatically improves governance outcomes; rather, they specify conditions under which AI-enabled coordination may enhance or undermine developmental capacity. Building on the mechanisms outlined above, the following theoretical propositions articulate how AI-driven governance capacities translate into economic growth within the digital developmentalism framework:

- **Proposition 1:** *We propose that higher levels of AI governance capacity enhance a government's ability to accelerate structural economic upgrading by improving coordination, information processing, and strategic policy alignment.*

Governments possessing advanced AI governance capacity, through data integration, algorithmic tools, and institutional learning, are better equipped to identify emerging industries, design targeted interventions, and stimulate technological innovation (Birkstedt et al., 2023). Such capacity enables faster transitions from low-to high-value sectors, consistent with the structural transformation logic of developmental theory.

- **Proposition 2:** *The model suggests that algorithmic coordination mediates the relationship between institutional quality and economic performance by translating governance capacity into more effective policy execution.*

Institutional quality provides the enabling environment for innovation, but algorithmic coordination determines its efficiency. AI-based coordination mechanisms operationalize institutional intent by improving information flows, monitoring compliance, and aligning incentives across sectors (Medaglia et al., 2024). Thus, the productivity of institutions increasingly depends on their algorithmic adaptability.

- **Proposition 3:** *We theorize that public data infrastructure moderates the relationship between AI governance capacity and economic growth, such that the developmental effects of AI-enabled governance are stronger in contexts with robust digital and data infrastructures.*

The relationship between AI governance and growth is contingent on the availability and quality of digital infrastructure. Nations with robust data architectures and secure interoperability frameworks experience stronger developmental dividends from AI adoption (World Bank, 2024). Conversely, weak digital foundations constrain the scaling of AI governance, limiting its macroeconomic impact.

- **Proposition 4:** *We propose that synergistic public-private digital ecosystems foster innovation-led growth by aligning state-led development objectives with private sector capabilities.*

Collaboration within digital ecosystems amplifies the growth effects of AI governance by combining public missions with private capabilities (Mazzucato, 2021). Through joint R&D, shared data platforms, and innovation incentives, such synergies accelerate technological diffusion, enhance productivity, and create inclusive developmental outcomes.

- **Proposition 5:** *We propose that AI-enhanced policy feedback increases governance adaptability by enabling faster learning and adjustment, while institutional resilience depends on whether such adaptive capacity is*

**Table 1**  
Core theoretical anchors and their contributions to the digital developmentalism framework.

| Theoretical Lens           | Key Authors                                            | How They Contribute             |
|----------------------------|--------------------------------------------------------|---------------------------------|
| Developmental State Theory | (Amsden, 1989; Evans, 1995; Johnson, 1982; Wade, 2018) | Foundation for state-led growth |
| Institutional Economics    | (Acemoglu & Robinson, 2013; North, 1990)               | Institutions as growth enablers |
| GPT Theory                 | Brynjolfsson and McAfee (2014)                         | AI as transformative technology |
| Digital Governance         | (de Jong et al., 2025; Van Noordt & Misuraca, 2022)    | Data-driven public management   |
| Public Value and AI Ethics | (Bryson et al., 2014; Moore, 2021; Wirtz et al., 2019) | Ethical governance dimension    |

embedded within robust legal, organizational, and accountability structures. In the absence of these safeguards, highly adaptive systems may remain vulnerable to brittleness and systemic failure.

Continuous policy learning enabled by AI allows governments to manage uncertainty and maintain stability in rapidly changing environments (UNDP, 2024). This adaptability becomes a key driver of resilience, ensuring that growth remains sustainable amid technological and global shocks.

These propositions collectively formalize the causal pathways through which digital developmentalism operates, linking AI governance capacity, coordination mechanisms, and infrastructure quality to innovation and growth outcomes.

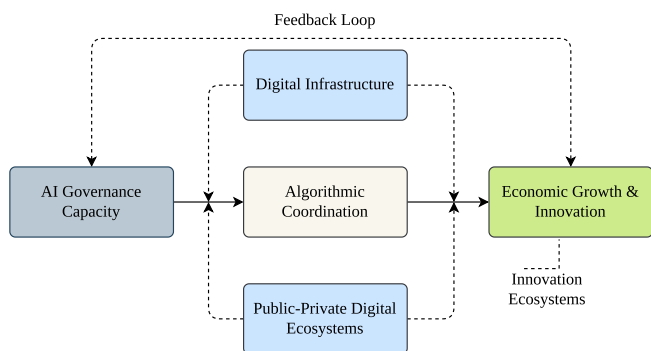
#### 4.4. Conceptual model

Drawing on the theoretical propositions, Fig. 1 presents a conceptual representation of hypothesized interactions among AI governance capacity, algorithmic coordination, and economic outcomes. The relationships illustrated in the model are analytically derived from existing theoretical literatures and are intended to capture theoretically plausible mechanisms rather than empirically validated causal dynamics. As such, the figure should be interpreted as a heuristic framework that organizes expectations about how AI-enabled governance may operate in practice, providing a basis for future empirical testing and comparative analysis rather than a depiction of confirmed relationships. At the core of the model lies AIGC, which directly influences Algorithmic Coordination, the primary mechanism through which governments manage complex economic systems. Algorithmic coordination, in turn, enhances economic growth and innovation performance, with this relationship being moderated by two contextual factors: public data infrastructure and public-private digital ecosystems. Additionally, AI-enhanced policy feedback serves as a reinforcing loop that improves both governance adaptability and coordination efficiency.

The model captures the dynamic feedback nature of AI-driven governance: as governments expand data infrastructures and algorithmic tools, their capacity for coordination and learning improves, leading to enhanced growth performance. In return, economic expansion generates new data and innovation opportunities, feeding back into the governance cycle. This reciprocal relationship positions AI not merely as a technological factor but as a structural force of governance evolution, redefining how states pursue development in the digital age.

### 5. Theoretical integration and discussion

This section situates the proposed framework within broader debates on state capacity, development, and governance, and discusses its theoretical implications.



**Fig. 1.** Conceptual framework of digital developmentalism (AI-governance-growth nexus).

#### 5.1. Comparison with the classical developmental state

Digital developmentalism builds on the intellectual foundation of the classical developmental state but reconfigures its operational logic through data, algorithms, and digital infrastructures. While both paradigms share the central principle of state-led coordination toward national development goals, they differ profoundly in their tools, informational environments, and adaptive capacities. Table 2 outlines these contrasts.

In the classical model, as theorized by Johnson (1982) and Evans (1995), developmental effectiveness was rooted in bureaucratic discipline, meritocratic civil services, and selective industrial policies. Coordination was hierarchical, guided by long-term industrial planning and negotiation between state agencies and large firms. Information flows were slow and fragmented, and policy evaluation occurred periodically through industrial output and export indicators. While this model achieved remarkable industrialization across East Asia, it was constrained by information lag, sectoral rigidity, and limited adaptability to non-linear global dynamics (Amsden, 1989; Wade, 2018).

In contrast, digital developmentalism retains the mission-oriented character of the developmental state but transforms its operational dynamics through algorithmic coordination and data-driven governance. Real-time data and predictive analytics replace industrial statistics as the informational foundation for decision-making. The state's role shifts from hierarchical command to network orchestration, coordinating public and private actors through digital platforms and interoperable information systems (Ahn & Chen, 2022). This transformation enhances precision, speed, and responsiveness in policy interventions. Moreover, continuous learning, enabled by AI-enhanced feedback loops, allows governments to recalibrate strategies dynamically rather than adhering to rigid industrial plans.

Thus, digital developmentalism combines the strategic intentionality of the classical state with the adaptive intelligence of a digital ecosystem. The developmental mission remains central, directing innovation, upgrading industries, and enhancing competitiveness, but the mechanisms of delivery evolve from bureaucratic to computational. The resulting model is not a replacement but an evolution: a computational extension of the developmental state's strategic logic suited to an era of complex, data-intensive economies. Fig. 2 illustrates the cyclical interaction between policy learning, AI governance capacity, algorithmic coordination, and economic outcomes, emphasizing continuous adaptation and feedback at the heart of digital developmentalism.

#### 5.2. Ethical, political, and institutional considerations

It is important to distinguish between the analytical and normative dimensions of the discussion. Analytically, the framework explains how AI-enabled governance systems may enhance coordination capacity, information processing, and policy adaptability, regardless of regime type or normative orientation. Normative considerations, such as concerns about democratic accountability, transparency, and the risk of "development without democracy", do not constitute explanatory claims about system performance, but rather evaluative judgments about the desirability and legitimacy of different governance trajectories. By separating these dimensions, the analysis clarifies how digital

**Table 2**

Comparative dimensions of the classical developmental state and digital developmentalism.

| Dimension              | Classical Developmental State | Digital Developmentalism |
|------------------------|-------------------------------|--------------------------|
| Coordination mechanism | Bureaucratic hierarchy        | Algorithmic coordination |
| Information base       | Industrial statistics         | Real-time data           |
| Core assets            | Capital, labor                | Data, AI infrastructure  |
| Policy feedback        | Periodic evaluation           | Continuous learning      |

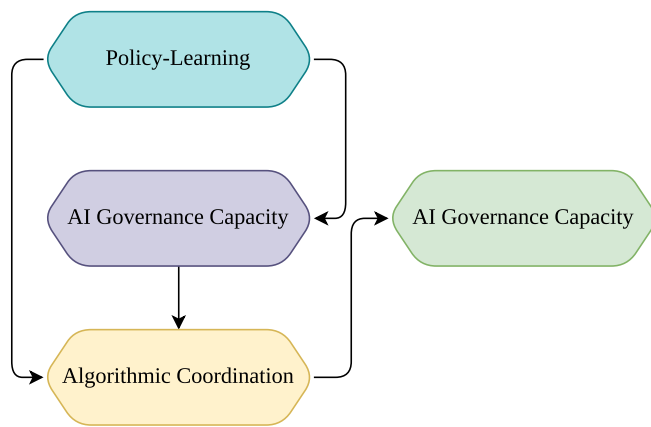


Fig. 2. Dynamic feedback loop of AI-Driven governance and economic outcome.

developmentalism functions as an explanatory framework, while ethical and democratic concerns are treated as normative implications that arise from, rather than define, the model.

While digital developmentalism promises efficiency and innovation, it also introduces ethical, political, and institutional dilemmas that complicate its implementation. The integration of AI into governance systems raises concerns about algorithmic opacity, where decision processes become inscrutable to citizens and even policymakers. When algorithms determine welfare eligibility, policing priorities, or credit allocation, the lack of transparency can erode public trust and accountability (Cows et al., 2023). Similarly, data-driven governance can amplify surveillance capabilities, increasing the risk of technocratic control over citizens. Without robust institutional safeguards, the digital state may drift toward authoritarian efficiency, optimizing for control rather than inclusion (Balarabe, 2025).

The “development without democracy” dilemma epitomizes this tension. Historically, developmental states justified strong bureaucratic autonomy as necessary for rapid industrialization. In the digital age, this rationale risks being replicated through technological systems that centralize decision power in opaque algorithmic infrastructures. AI can enhance administrative capacity but also concentrate informational asymmetries, allowing states to govern through data extraction and behavioral prediction. Such dynamics may increase efficiency but reduce democratic participation, reinforcing what some scholars call algorithmic authoritarianism (Filgueiras, 2022). Hence, the challenge for digital developmentalism is not merely technical but normative: how to harness AI for developmental ends while preserving transparency, accountability, and rights-based governance.

An additional strength of the proposed framework lies in its explicit integration of ethical and institutional considerations into the analysis of AI-enabled governance. Rather than presenting a purely positive or technocratic account of digital transformation, the model incorporates risks such as surveillance expansion, algorithmic opacity, and the tension between efficiency and democratic accountability often described as “development without democracy.” By addressing these concerns alongside its analytical mechanisms, the framework offers a more comprehensive account of digital developmentalism that recognizes both its developmental potential and its institutional vulnerabilities. This dual focus enhances the framework’s relevance for both explanatory analysis and responsible governance debates.

### 5.3. Policy and global implications

The primary policy implication of the digital developmentalism framework concerns how governments can leverage AI-enabled governance to support structural transformation and long-term economic growth. Rather than treating digitalization as an efficiency-enhancing

add-on, the framework emphasizes the role of AI governance capacity in coordinating investment, guiding innovation, and enabling adaptive industrial policy (World Bank, 2024). From a developmental perspective, this implies that public investment in data infrastructure, algorithmic capabilities, and institutional learning should be viewed as core components of modern development strategy rather than auxiliary administrative reforms.

For developing and middle-income economies, digital developmentalism highlights the importance of sequencing and capacity building. Effective use of AI in governance presupposes foundational investments in digitized records, interoperable public systems, and regulatory capacity. Without these prerequisites, AI-driven policy tools risk remaining symbolic or unevenly applied. Developmental policy therefore requires a staged approach in which digital infrastructure, human capital, and institutional readiness evolve alongside more advanced AI applications.

At the global governance level, digital developmentalism also reshapes the political economy of interdependence. Countries with strong digital capabilities gain strategic advantages in data trade, cross-border AI regulation, and technological standard-setting. Conversely, weaker states risk digital dependency, relying on foreign platforms for essential governance functions. Addressing these asymmetries requires multilateral frameworks that promote equitable access to digital infrastructure and capacity-sharing mechanisms for AI governance. Initiatives such as the UNDP’s “AI for Sustainable Development” and the OECD’s AI Observatory exemplify early steps toward collective digital governance. Nevertheless, achieving inclusive digital development will depend on whether global institutions can reconcile innovation with fairness in the emerging digital order.

### 5.4. Theoretical contribution summary

The framework of digital developmentalism contributes to governance and development theory in four interrelated ways. First, it positions AI as a new form of state capability. Whereas previous eras emphasized fiscal or bureaucratic capacity, the digital era emphasizes computational capacity, the ability to govern through data and algorithms. This reconceptualization extends theories of state capacity (Acemoglu & Robinson, 2013; Evans, 1995) by integrating the technological dimension of governance. It reframes the state as an information processor, capable of converting data into developmental intelligence.

Second, digital developmentalism bridges institutional, technological, and governance literatures. It synthesizes institutional economics’ focus on rules and incentives, innovation studies’ emphasis on technological change, and governance theory’s concern with policy effectiveness. Third, the theory provides an analytical framework for policy and empirical research. The conceptual model proposed in Section 4 delineates testable relationships between AI governance capacity, algorithmic coordination, and economic performance. These propositions can inform empirical studies using cross-country indicators of AI readiness, governance quality, and innovation output (World Bank, 2024).

Finally, digital developmentalism advances a normative vision of inclusive technological governance. It argues that development in the AI era must balance efficiency with ethics, innovation with inclusion, and sovereignty with interdependence. Governments must govern not only with intelligence but also for collective well-being. The theoretical synthesis presented here therefore repositions governance as both a driver of growth and a guardian of fairness in the digital age. By conceptualizing AI as an endogenous element of state capability and a transformative force of economic coordination, digital developmentalism sets the stage for reimagining development theory for the twenty-first century, one that recognizes that the future of growth is as much about algorithms and data as it is about institutions and policy.

The theoretical framework rests on several analytical assumptions that define its scope of inference. First, it assumes the presence of a minimally functional state apparatus capable of coordinating policy and

managing digital infrastructures. Second, it presupposes access to basic digital and data infrastructures that enable algorithmic coordination and policy learning, even if such capacities are uneven or evolving. Third, the framework assumes that AI governance is embedded within institutional contexts that allow for policy adaptation and feedback rather than purely symbolic or ceremonial adoption. These assumptions imply clear limitations: the framework is less applicable to contexts characterized by severe state fragility, extreme data scarcity, or the absence of institutional learning mechanisms. Accordingly, digital developmentalism should be understood as a conditional theory of governance transformation, whose explanatory power varies with institutional maturity and digital readiness.

## 6. Implications and future research

This section outlines the theoretical and policy implications of the framework and identifies priorities for future empirical research.

### 6.1. Theoretical implications

The theory of digital developmentalism makes two major contributions to the academic understanding of governance and economic development. First, it redefines the concept of state capacity to include algorithmic and data capabilities. Traditional formulations of state capacity emphasized fiscal strength, bureaucratic professionalism, and regulatory quality (Acemoglu & Robinson, 2013; Evans, 1995). In contrast, digital developmentalism conceptualizes AI governance capacity as a distinct dimension of state capability, comprising data infrastructures, machine learning systems, and institutional learning processes that collectively enhance the precision and adaptability of public action. These reframing positions computational intelligence as a critical source of developmental advantage in the twenty-first century, analogous to the role of bureaucratic competence in the industrial era.

Second, the framework advances an AI-based theory of institutional evolution in development economics. Building on North's (1990) and Nelson and Winter's (1982) evolutionary perspectives, it argues that institutions evolve in response to technological paradigms, and AI represents a transformative general-purpose technology that compels states to reorganize their decision structures. Digital developmentalism thus extends institutional theory by integrating technological endogeneity, recognizing that institutions are not only shaped by technology but also co-evolve with it. This insight opens a theoretical pathway for understanding how AI governance fosters dynamic capabilities, enabling states to adapt continuously to economic complexity and global uncertainty.

Furthermore, by synthesizing elements of GPT theory and governance studies, digital developmentalism bridges the micro–macro divide in development theory. It links the micro-level processes of algorithmic governance, such as policy analytics and data feedback loops, to macro-level outcomes like productivity, innovation diffusion, and inclusive growth. In doing so, it provides a unified theoretical model that connects technological transformation with the institutional logic of economic development.

### 6.2. Policy implications

From a policy standpoint, digital developmentalism offers practical guidance for governments seeking to harness AI for sustainable and inclusive growth. The first implication concerns the design of AI-driven development strategies. Governments must view AI not simply as a tool for automation but as a strategic governance capability, one that enables evidence-based policymaking, predictive planning, and innovation coordination. This requires national strategies that integrate AI deployment with long-term development missions such as industrial upgrading, digital inclusion, and environmental sustainability (UNDP, 2024). For instance, countries like Singapore and South Korea have

embedded AI initiatives within their national innovation frameworks, linking public investment in digital infrastructure to specific industrial outcomes (Joo, 2023).

The second policy implication is the need for ethical, inclusive, and transparent digital ecosystems. As digital developmentalism amplifies state power through data and algorithms, governments must ensure that these technologies are governed responsibly. Ethical AI governance, guided by transparency, accountability, and fairness, is essential to prevent discrimination, exclusion, and misuse of data (COWLS et al., 2023). Institutional frameworks should include independent oversight bodies, algorithmic audit mechanisms, and participatory design processes to safeguard public trust. In developing economies, this is particularly crucial to avoid reinforcing structural inequalities and to ensure that AI-driven growth contributes to social well-being rather than elite capture.

Third, the theory underscores the importance of public–private collaboration in digital governance. Effective AI-driven development requires synergy between state leadership and market innovation. Governments should cultivate partnerships with technology firms, research institutions, and startups through open data platforms, digital sandboxes, and joint R&D programs. Such collaborations can accelerate innovation diffusion while aligning private incentives with public goals (Mazzucato, 2021). However, they must be governed by transparent frameworks that prevent monopolistic dominance and ensure equitable value distribution across the digital economy.

Finally, digital developmentalism has geopolitical implications. The competition between models of AI governance, such as China's state-led developmentalism versus the European Union's human-centered approach, illustrates that digital governance is not value-neutral (Dragu & Lupu, 2021; OECD, 2023). Policymakers must navigate this landscape by developing context-specific frameworks that balance efficiency, autonomy, and rights. For developing nations, the priority should be to build sovereign digital capacity, reducing dependence on foreign data infrastructures while integrating globally accepted standards for ethical AI deployment. In this sense, digital developmentalism is both a national and global project, requiring coordination across policy domains, technological regimes, and governance levels.

### 6.3. Limitations

This study is subject to several limitations inherent to theory-building research. First, the relationships depicted in the conceptual model are analytically derived and remain hypothetical rather than empirically tested. The framework articulates theoretically plausible governance mechanisms but does not establish demonstrated causal effects. Empirical validation is therefore necessary to assess the magnitude, robustness, and contextual variability of the proposed relationships.

Second, the generalizability of the framework cannot be assumed across all national or institutional settings. Its explanatory power is likely conditioned by levels of digital infrastructure, administrative competence, and governance maturity. The model should therefore be understood as a conditional theory rather than a universally applicable template. Third, the measurement of key constructs presents methodological challenges. Concepts such as AI governance capacity, algorithmic coordination, and public–private digital synergy are multidimensional and analytically rich, making them difficult to operationalize through single indicators or standardized metrics. Future empirical applications will require careful construct specification and contextual adaptation.

Fourth, the framework is inherently context dependent and temporally situated. Its mechanisms may operate differently across political systems, regulatory traditions, and stages of digital maturity, and ongoing technological evolution may require future refinement of the model. Fifth, data availability and transparency constraints limit short-term empirical testability, as many governments do not disclose granular information on AI systems and algorithmic governance practices.

Sixth, the theory presupposes a baseline level of administrative digitization and institutional competence, creating the risk of a capability trap that may widen developmental divides.

Last but not the least, the framework does not fully theorize the legal determinants of digital developmentalism, including intellectual property regimes, data sovereignty laws, and administrative law reforms, which are essential enabling conditions for AI-enabled governance.

#### 6.4. Future research directions

Although the present study is conceptual, the proposed framework lends itself to future empirical operationalization. The theoretical model of digital developmentalism opens several promising avenues for empirical and comparative research. First, future studies should empirically test the relationship between AI governance capacity and economic performance using quantitative indicators. Cross-country analyses could employ composite indices such as the OECD AI Policy Observatory metrics, World Bank Governance Indicators, and AI readiness scores to examine whether higher AIGC correlates with greater innovation output, productivity, or GDP growth (World Bank, 2024). Structural equation modeling or panel regression techniques could help validate the mediating role of algorithmic coordination and the moderating effect of digital infrastructure on growth outcomes.

Second, comparative case studies are essential to contextualize how digital developmentalism manifests across governance systems. Countries such as Singapore, Estonia, and China offer contrasting trajectories: Singapore's technocratic governance combines efficiency with inclusiveness; Estonia exemplifies a small-state model of interoperable digital infrastructure; and China represents a high-capacity, centralized approach emphasizing technological self-reliance (Penmetsa & Bruque Camara, 2022). Comparing these models could yield insights into the institutional prerequisites, policy instruments, and ethical trade-offs that shape digital developmental outcomes.

Third, future research should explore quantitative modeling of AI-governance-growth linkages using dynamic systems or complexity-based simulations. Such models can help explain feedback effects between AI adoption, policy adaptability, and innovation diffusion. For instance, system dynamics modeling could illustrate how investments in data infrastructure and algorithmic tools generate long-term growth multipliers through improved coordination efficiency.

Fourth, scholars may investigate the distributional and ethical dimensions of digital developmentalism. How does AI-driven governance affect inequality, regional disparity, or political legitimacy? To what extent can ethical AI frameworks mitigate risks of surveillance, bias, or exclusion in developmental contexts? These questions are critical for ensuring that digital transformation remains aligned with human-centered and sustainable development goals.

Finally, interdisciplinary collaboration between economists, political scientists, technologists, and ethicists will be crucial to refine this theoretical model. As AI continues to evolve, so too must the frameworks that explain its governance implications. Empirical inquiry, policy experimentation, and comparative analysis can jointly advance digital developmentalism from a conceptual construct to a robust theory of 21st-century state-led growth.

## 7. Conclusion

This study advances digital developmentalism as a novel framework for understanding the evolving relationship between artificial intelligence, governance, and economic growth. By reconceptualizing state capacity in computational terms, it shifts attention from traditional bureaucratic hierarchy toward algorithmic coordination as a defining feature of contemporary governance. In doing so, the paper extends developmental state theory into the digital era while integrating institutional economics, general-purpose technology theory, and digital governance research into a coherent explanatory architecture. The

result is a structured, mechanism-based model that clarifies how AI-enabled coordination, data infrastructures, and feedback systems may reshape economic steering and structural transformation.

The core contribution lies in identifying algorithmic coordination as a qualitative transformation in the logic of the state. Where earlier models emphasized hierarchical planning and administrative command, digital developmentalism conceptualizes governance as data-intensive, predictive, and adaptive. State capacity, in this view, rests increasingly on computational intelligence: the ability to process information at scale, learn from real-time feedback, and align public and private actors within digital ecosystems. This reframing broadens traditional accounts of institutional capability and situates technological change at the center of developmental analysis. Simultaneously, the framework remains analytically grounded rather than prescriptive. Its propositions are theoretical and contingent, intended to guide empirical inquiry rather than assert universal convergence. By combining conceptual clarity with interdisciplinary integration, digital developmentalism offers a foundation for future research into how AI may transform, and potentially redefine, the developmental role of the state.

#### Data availability statement

This study does not involve the collection, generation, or analysis of empirical data. The findings are based on theoretical synthesis and publicly available scholarly sources.

#### Ethical statement

This study is a theoretical and conceptual analysis and therefore did not involve human participants, personal data collection, or experimental interventions. All secondary materials, including published literature and official policy documents, were used responsibly and with full acknowledgment of original sources. The research adheres to the principles of academic integrity, transparency, and ethical scholarship as outlined in the Declaration of Helsinki (2013) and relevant institutional research ethics guidelines. The author affirms that no conflicts of interest, plagiarism, or unethical research practices are associated with this work, and all ideas are presented with respect for intellectual property and responsible knowledge dissemination.

#### Declaration of the use of generative AI

During the preparation of this manuscript, the author used QuillBot Premium and ChatGPT 5.2 to enhance language clarity and improve overall manuscript quality. After using these tools, the author carefully reviewed, revised, and edited the content as necessary and takes full responsibility for the final version of the publication.

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#### Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Syed Asad Abbas Bokhari reports administrative support and article publishing charges were provided by Nazarbayev University Graduate School of Public Policy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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